DUST on the K=32 code-carried model: the 500-step $\sigma\times$ `dust_lr_scale` grid

Research note

2026-10-22.

Single runs, seed 1337; the encoder arm and the baseline trained on new_york (8× RTX 5090), the backprop arm on georgia (8× A100).

1 Setup

The model is the K=32 code-carried hard Top-32 model of `docs/kj-vs-hard-topk-code-residual.tex` (`configs/cr_kj/ma_cr_hard_k32_pnorm.yaml`): 8 blocks of width 768, abs-TopK over $N = 1536$ features at `residual_out` of every block, the code carried between blocks, post-norm, magnitude–direction (MD) decoupling, batch 96×512 tokens, a 20k-step cosine schedule with 500 warmup steps.

Every DUST run (`configs/cr_dust/hp500/`, method in `docs/dust-topk.md`) adds the DUST bottleneck optimization: the $J = 480$ pool receives the noise, population 128 (16 draws per bottleneck), schedule $[1, 1]$ (every other optimizer step a DUST step), $\gamma = 0$, gradient scale 1, a separate optimizer for the DUST steps sharing the MD parameterization, and no gradient clipping. The grid is `dust_target` $\in \{\text{encoder}, \text{backprop}\} \times \sigma \in \{0.03, 0.1, 0.3, 0.9\} \times$ `dust_lr_scale` $\in \{1, 10, 30, 100, 300\}$. Each run trains 500 steps of the unchanged 20k-step schedule, so the window lies inside the LR warmup ($1.2 \cdot 10^{-6}$ to $6 \cdot 10^{-4}$; the DUST steps use that value times `dust_lr_scale`), with validation every 50 steps on 40 batches of 24 windows.

The baseline is the same model trained with the same settings and no DUST steps (`ny_hp500_hard_k32`); the original madrid run of the model (gradient clipping at 1.0) is quoted for reference.

2 Encoder target

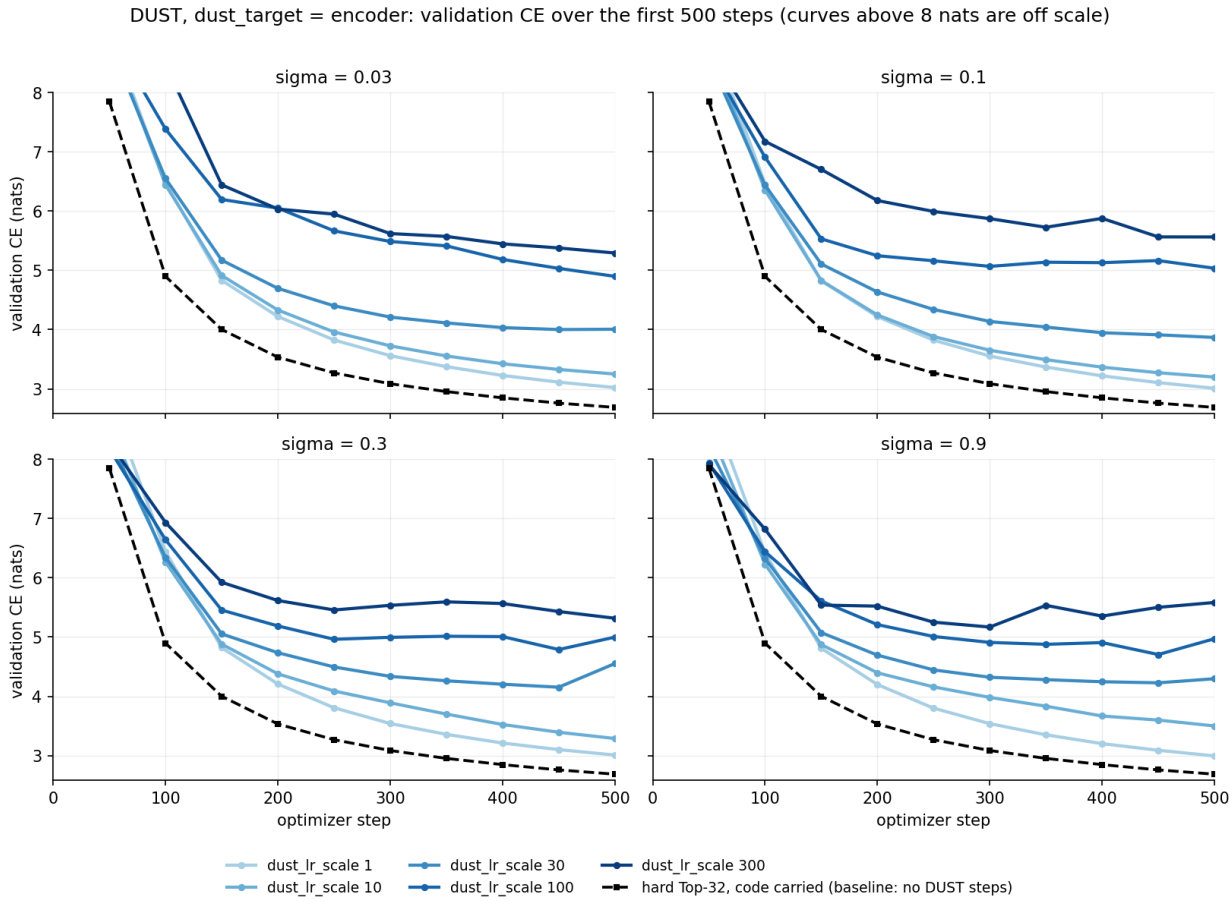


Figure 1: `dust_target = encoder`: validation CE over the first 500 steps, one panel per σ, one curve per `dust_lr_scale` (one hue, light to dark with the multiplier); dashed black, the hard Top-32 baseline. Curves above 8 nats are off scale.

Validation CE at step 500 (bold: the best cell):

σ \ `dust_lr_scale`	1	10	30	100	300
0.03	3.0220	3.2500	4.0055	4.8971	5.2906
0.1	3.0096	3.1989	3.8673	5.0334	5.5611
0.3	3.0101	3.2891	4.5535	4.9989	5.3168
0.9	**2.9957**	3.5009	4.2972	4.9715	5.5825
hard Top-32 baseline (no DUST steps, same settings)	2.6875				
hard Top-32, the original madrid run (clipping 1.0)	2.6846				

3 Backprop target

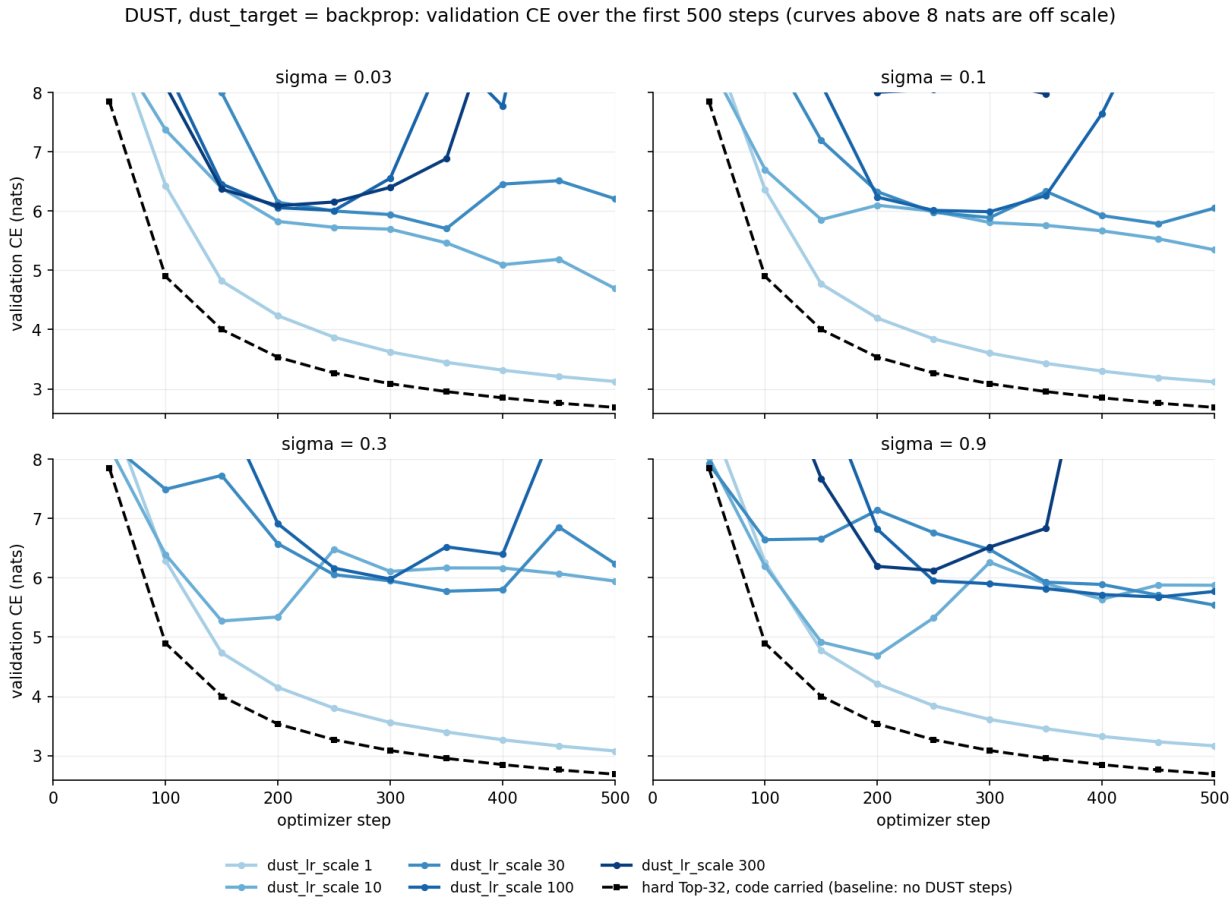


Figure 2: `dust_target = backprop`: the same layout. Curves above 8 nats are off scale.

Validation CE at step 500:

σ \ `dust_lr_scale`	1	10	30	100	300
0.03	3.1246	4.6930	6.2055	13.0487	9.7345
0.1	3.1169	5.3437	6.0478	8.8164	9.9774
0.3	**3.0794**	5.9428	6.2382	11.9248	9.6194
0.9	3.1646	5.8732	5.5386	5.7687	9.2661
hard Top-32 baseline (no DUST steps, same settings)	2.6875				
hard Top-32, the original madrid run (clipping 1.0)	2.6846				

4 Results

No cell reaches the baseline. In both arms the validation CE at step 500 rises monotonically with `dust_lr_scale`, and the multiplier 1 is the best value at every σ: 2.996 to 3.022 in the encoder arm, 0.31 to 0.33 nats above the baseline, and 3.08 to 3.16 in the backprop arm. The dependence on σ is weak at the multiplier 1 (a spread of 0.03 nats in the encoder arm and 0.09 in the backprop arm) and does not change the ordering by the multiplier anywhere.

The two targets differ in how the larger multipliers fail. In the encoder arm the curves keep descending at every setting, only more slowly and from a higher level as the multiplier grows; at 100 and 300 they plateau near 5 to 5.6 nats within the window. In the backprop arm, where the

estimate is backpropagated to every parameter before the gate inputs, every multiplier from 10 on stalls near 5 to 6 nats, and 100 and 300 diverge at every σ (final values of 5.8 to 13 nats), with the pre-activation scale growing by orders of magnitude within the first tens of steps.

Within the first 500 steps the DUST steps therefore cost against hard steps at every setting of the grid, least when the DUST learning rate is the schedule's own, and the encoder target is less harmful than the backprop target at every σ and multiplier.

5 Sigma sweep at population 1024

`dust_lr_scale` 1, population 1024 (128 draws per bottleneck, eight times the grid's), both targets, the four σ values, otherwise the grid's settings; all eight runs on new_york.

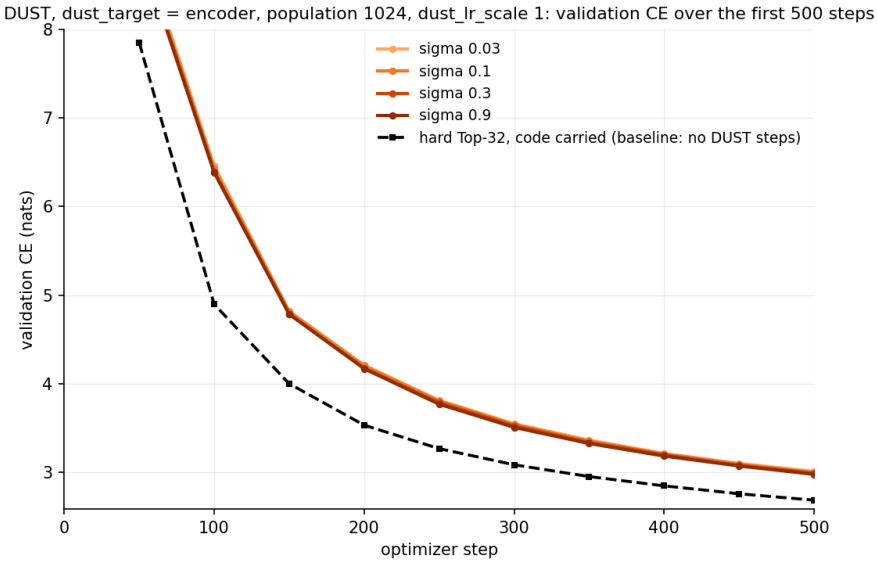


Figure 3: Population 1024, `dust_target = encoder`: validation CE over the first 500 steps, one curve per σ (one hue, light to dark with σ); dashed black, the hard Top-32 baseline.

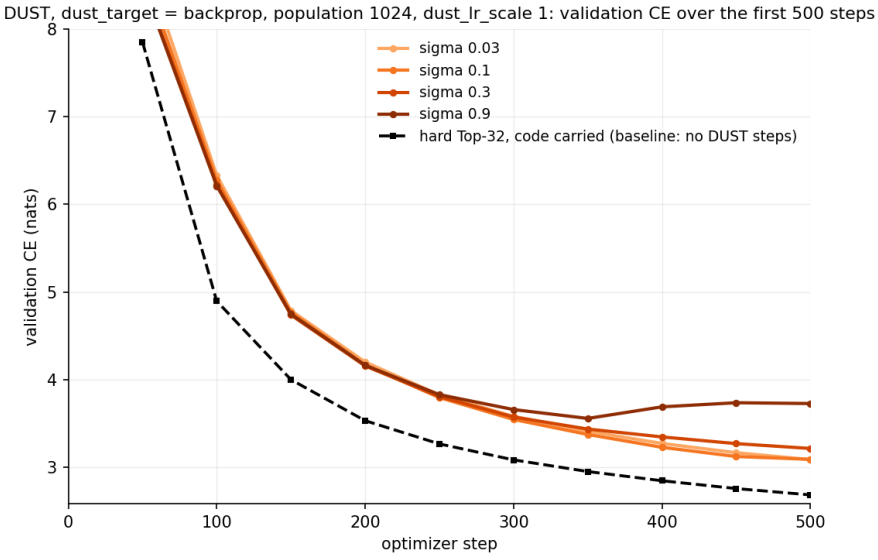


Figure 4: Population 1024, `dust_target = backprop`: the same layout.

Validation CE at step 500 (population 128 at `dust_lr_scale` 1 in parentheses):

dust_target \ σ	0.03	0.1	0.3	0.9
encoder	3.0117 (3.0220)	2.9975 (3.0096)	2.9923 (3.0101)	**2.9790** (2
backprop	3.0896 (3.1246)	3.0930 (3.1169)	3.2166 (3.0794)	3.7286 (3.
hard Top-32 baseline (no DUST steps, same settings)	2.6875			
hard Top-32, the original madrid run (clipping 1.0)	2.6846			

Eight times the population lowers the encoder arm's CE at step 500 by 0.010 to 0.018 nats at every σ, and the four curves are indistinguishable until the last 100 steps, where they order by σ with the largest best. The backprop arm improves by 0.02 to 0.04 at $\sigma = 0.03$ and 0.1 and worsens at 0.3 and 0.9, where the curves turn upward after step 350 ($\sigma = 0.9$ ends at 3.73). The gap to the baseline, 0.29 nats in the encoder arm at best, is therefore not a matter of the estimate's variance at this population: the estimate's variance is eight times smaller and the gap moved by a few hundredths.

6 Without MD decoupling

Pending: the hard Top-32 run without MD decoupling (`model.decouple = false` with `model.md_init = true`, so the initialization is identical and only the optimizer changes: ordinary AdamW with weight decay 0.1, the convention of `docs/md-init-vs-decoupling.tex`), the population-1024 σ sweep without decoupling at $\gamma = 0$ (georgia) and at $\gamma = 0.99$ (new_york). To be added: the hard run with and without decoupling; and, per γ, a two-panel figure (encoder left, backprop right) of the sweep against the hard run without decoupling, with its table.

7 Runs on Drive

- Encoder arm, new_york: `gdrive:weight-sparsity/runs_new_york/ny_hp500_enc_sg<s>_lr<l>` for `<s>` $\in$ {`0p03`, `0p1`, `0p3`, `0p9`} and `<l>` $\in$ {1, 10, 30, 100, 300}, twenty runs.
- Baseline, new_york: `gdrive:weight-sparsity/runs_new_york/ny_hp500_hard_k32`.
- Population-1024 sweep, new_york: `gdrive:weight-sparsity/runs_new_york/ny_hp500_<a>_p1024_sg<s>_lr1` for `<a>` $\in$ {`enc`, `bp`}, eight runs.
- Backprop arm, georgia: `gdrive:weight-sparsity/runs_georgia/ga_hp500_bp_sg<s>_lr<l>`, the same twenty cells.
- Original hard Top-32 run (reference value only): `gdrive:weight-sparsity/runs_madrid/ma_cr_hard_k32_pnorm`.

Figures and tables: `scripts/plot_dust_hp500.py` over the `metrics.jsonl` of these runs. The Markdown version of this note, `docs/dust/hp500-sigma-lr.md`, lists every run name in full.